

SECR2043 OPERATING SYSTEMS

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Section : 02

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write **"No output"** instead.

1.	Display your current working directory.	
	Command	Output
	<code>pwd</code>	<code>woo@secr2043:~\$ pwd /home/woo</code>
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	<code>ls -a ~</code>	<code>woo@secr2043:~\$ ls -a ~bash_logout .bashrc .cache .profile .ssh</code>
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
	<code>mkdir ~/WOOCHENGSHUAN</code>	No output
4.	Change your current working directory to "os_lab".	
	Command	Output
	<code>cd ~/WOOCHENGSHUAN</code>	No output
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	Output

	vim hello.txt		
6.	Display the content of "hello.txt".		
	Command	Output	
	ls hello.txt	No output	
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.		
	Command	Output	
	cp hello.txt hello_copy.txt	No output	
8.	Rename "hello_copy.txt" to "hello_bak.txt".		
	Command	Output	
	Mv hello_copy.txt hello_bak.txt	No output	
9.	Delete "hello.txt".		
	Command	Output	
	rm hello.txt	No output	
10.	Display your home directory with additional information, together with subdirectories contents.		
	Command	Output	

<pre>ls -alR ~</pre>	<pre>woos@secr2043:~\$ ls -alR ~ /home/woo: total 44 drwxr-x--- 5 woo woo 4096 Apr 23 15:12 . drwxr-xr-x 6 root root 4096 Apr 22 16:03 .. -rw-r--r-- 1 woo woo 220 Apr 22 16:03 .bash_logout -rw-r--r-- 1 woo woo 3771 Apr 22 16:03 .bashrc drwx----- 2 woo woo 4096 Apr 22 16:12 .cache -rw----- 1 woo woo 12288 Apr 23 15:12 .hello.txt.swp -rw-r--r-- 1 woo woo 807 Apr 22 16:03 .profile drwx----- 2 woo woo 4096 Apr 22 16:13 .ssh drwxrwxr-x 2 woo woo 4096 Apr 23 15:18 WOOCHENGSHUAN /home/woo/.cache: total 8 drwx----- 2 woo woo 4096 Apr 22 16:12 . drwxr-x--- 5 woo woo 4096 Apr 23 15:12 .. -rw-r--r-- 1 woo woo 0 Apr 22 16:12 motd.legal-displayed /home/woo/.ssh: total 16 drwx----- 2 woo woo 4096 Apr 22 16:13 . drwxr-x--- 5 woo woo 4096 Apr 23 15:12 .. -rw----- 1 woo woo 978 Apr 22 16:13 known_hosts -rw-r--r-- 1 woo woo 142 Apr 22 16:12 known_hosts.old /home/woo/WOOCHENGSHUAN: total 36 drwxrwxr-x 2 woo woo 4096 Apr 23 15:18 . drwxr-x--- 5 woo woo 4096 Apr 23 15:12 .. -rw----- 1 woo woo 12288 Apr 23 15:18 .hello.txt.swn -rw----- 1 woo woo 4096 Apr 23 15:16 .hello.txt.swo -rw----- 1 woo woo 12288 Apr 23 15:07 .hello.txt.swp</pre>
Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
	touch hello.txt	No output
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
	echo "Hello World" > hello.txt	No output
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	echo "This is a test" >> hello.txt	No output
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	cat hello.txt	woo@secr2043:~\$ cat hello.txt Hello World This is a test
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	echo "Goodbye World" > bye.txt	No output
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	cat hello.txt bye.txt	woo@secr2043:~\$ cat hello.txt bye.txt Hello World This is a test Goodbye World
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output
	cat hello.txt bye.txt > greeting.txt	No output

8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	woo@secr2043:~\$ cat greeting.txt Hello World This is a test Goodbye World
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
	nano greeting.txt	GNU nano 7.2 Hello World This is a test Goodbye World
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	woo@secr2043:~\$ cat greeting.txt Hello woochengshuan This is a test Goodbye woochengshuan
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc.
For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
	nano hello.c	No output
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
	gcc hello.c -o hello	No output
3.	Run the executable file "hello" and display its output.	
	Command	Output
	./hello	woo@secr2043:~\$./hello Hello World
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
	nano sum.c	No output
5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output
	gcc -Wall sum.c -o sum	No output
6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
	./sum	woo@secr2043:~\$./sum Enter the first number: 10 Enter the second number: 20 Sum: 30
7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output
	./sum	woo@secr2043:~\$./sum Enter the first number: -5 Enter the second number: 15 Sum: 10

8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
	nano echo.c	No output
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
	gcc -Wall echo.c -o echo	No output
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output
	./echo	<pre> woo@secr2043:~\$./echo Input the text: Hello World Output the text: Hello World </pre>
	Total Mark:	